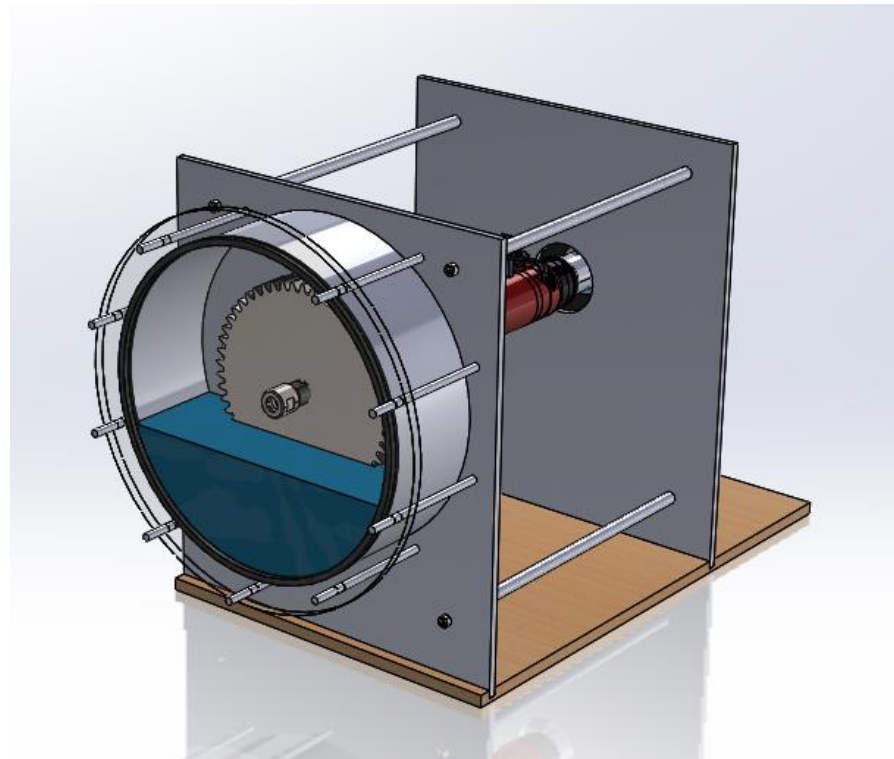


An Experimental Investigation of Oil Churning and Flow Regimes



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Background - Gearbox

- Numerous applications in the automotive industry
- Maximum efficiency of 98-99%
- Power loss for high power applications could still be in Megawatts
- A better understanding on the power loss would allow for more improved designs
- Power loss divided into 2 categories
 - load-dependent: Friction Loss, etc
 - load-independent: Oil Churning, Windage, etc



Need and Aim of Project

- Few experimental research was done on Oil Churning
- Existing formulas are widely different and situational
- Thus, a new research was commenced to validate existing formulas
- Expand knowledge on the relation between churning loss and flow regimes



Existing Formulas

Boness's[1] expression for Churning Torque C_{ch} :

$$C_{ch} = \frac{\rho}{2} \omega^2 S_m R_p^3 C_m$$

Where, ρ : Density of the Lubricant

ω : Rotation Speed

S_m : Submerged Surface Area

R_p : Pitch Radius of Gear

C_m : Dimensionless Torque

And the dimensionless torque depends on the flow regime:

For laminar flows($Re < 2000$)

$$C_m = \frac{20}{Re}$$

For intermediate flow regimes($2000 < Re < 100,000$)

$$C_m = 8.6 \times 10^{-4} Re^{1/3}$$

For Turbulent flows($100,000 < Re$)

$$C_m = \frac{5 \times 10^8}{Re^2}$$

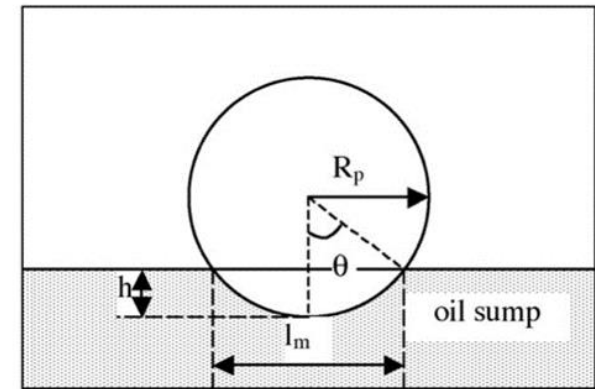
Existing Formulas Cont.

Changenet & Velez [2] in a preliminary research suggested the following definitions,
Reynold's Number:

$$Re = \frac{\omega R_p b}{\nu}$$

Froude's Number:

$$Fr = \frac{\omega^2 R_p}{g}$$



The follow-up research from Changenet et al[3] introduced the centrifugal acceleration of the lubricant γ ,

$$\gamma = \omega^2 (R_p b m)^{1/3}$$

Where, b : Width of Gear

m : Gear Module

ν : Kinetic Viscosity

Existing Formulas Cont.

Changenet et al[3] suggested the C_m can be expressed as:

if $\gamma < 750 \text{ m/s}^2$ and $Re < 4000$

$$C_m = 1.366 \left(\frac{h}{D_p} \right)^{0.45} \left(\frac{V_0}{D_p^3} \right)^{0.1} Fr^{-0.6} Re^{-0.21} \left(\frac{b}{R_p} \right)^{0.21} \quad (1)$$

if $\gamma < 750 \text{ m/s}^2$ and $Re > 4000$

$$C_m = 0.239 \left(\frac{h}{D_p} \right)^{0.45} \left(\frac{V_0}{D_p^3} \right)^{0.1} Fr^{-0.6} \left(\frac{b}{R_p} \right)^{0.21} \quad (2)$$

if $\gamma > 1250 \text{ m/s}^2$ and $Re < 4000$

$$C_m = 20.8 \left(\frac{h}{D_p} \right)^{0.1} \left(\frac{V_0}{D_p^3} \right)^{-0.35} Fr^{-0.88} Re^{-0.21} \left(\frac{b}{D_p} \right)^{0.85} \quad (3)$$

if $\gamma > 1250 \text{ m/s}^2$ and $Re > 4000$

$$C_m = 3.644 \left(\frac{h}{D_p} \right)^{0.1} \left(\frac{V_0}{D_p^3} \right)^{-0.35} Fr^{-0.88} \left(\frac{b}{D_p} \right)^{0.85} \quad (4)$$

- Dimensional analysis

Using Changenet & Velez's[2] equations as example

$$C_{ch} = f(m, h, D_p, b, \mu, \rho, h, V_0, g, \Omega)$$

$$C_m = \psi_1 \left(\frac{m}{D_p} \right)^{\psi_2} \left(\frac{b}{D_p} \right)^{\psi_3} \left(\frac{h}{D_p} \right)^{\psi_4} \left(\frac{V_0}{D_p^3} \right)^{\psi_5} Re^{\psi_6} Fr^{\psi_7}$$

- Coefficients found by experiments

For example, only changing the Reynold's number by heating the oil up and changes the viscosity

$$C_m \propto Re^{\psi_6}$$

By comparing the different in C_m produced, the influence of the variable could then be found

$$C_{m1} = k \times Re_1^{\psi_6}$$

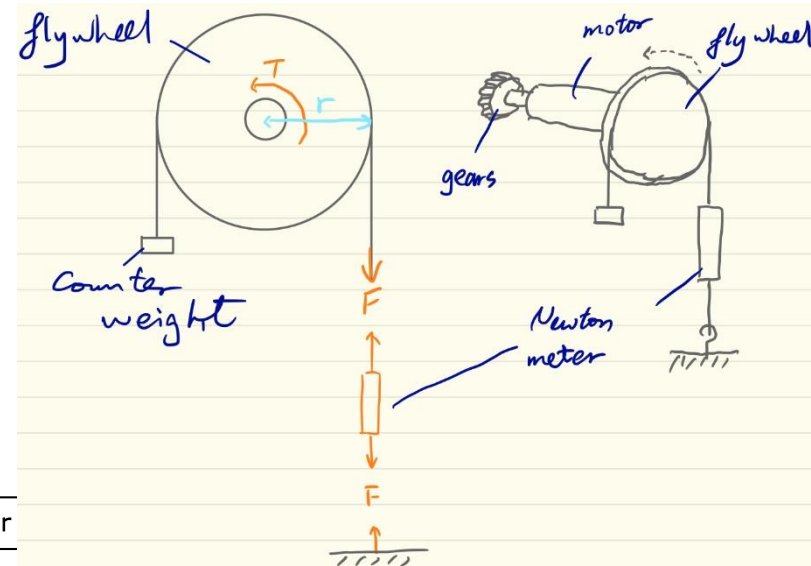
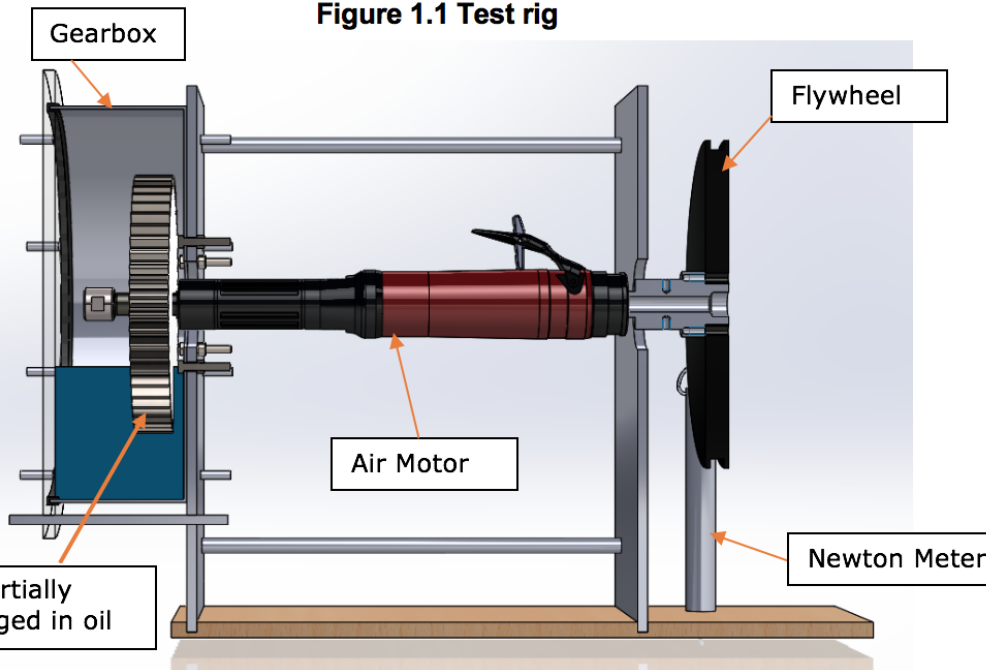
$$C_{m2} = k \times Re_2^{\psi_6}$$

Test Rig

- Rotational speed controlled by pressure and measured by tachometer
- “Recoil” Churning Torque measured by Newton meter
- Repeat without oil to extract Churning Torque

$$P_{ch} = P_{Total} - P_{Windage+Internal}$$

Figure 1.1 Test rig



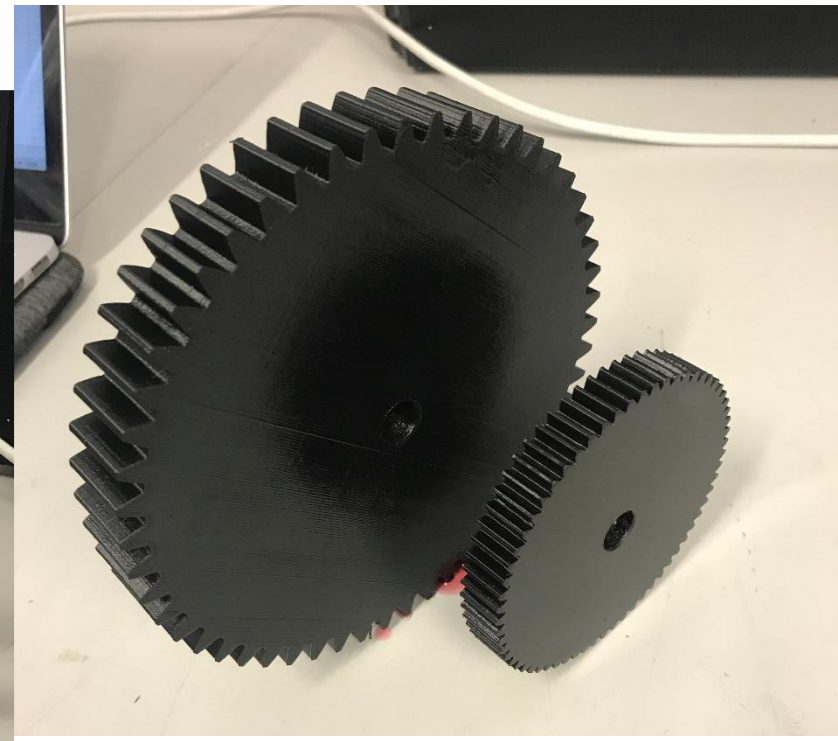


Testing Parameters

Experiments conducted on 2 different gears, 1 oil and 2 different immersion depth overall

Table 1 Gear data

	Pitch Diameter(mm)	Module(mm)	Face Width(mm)	Number of teeth
Gear 1	96	1.5	14	64
Gear 2	159	3	24	53



Testing Parameters Cont.

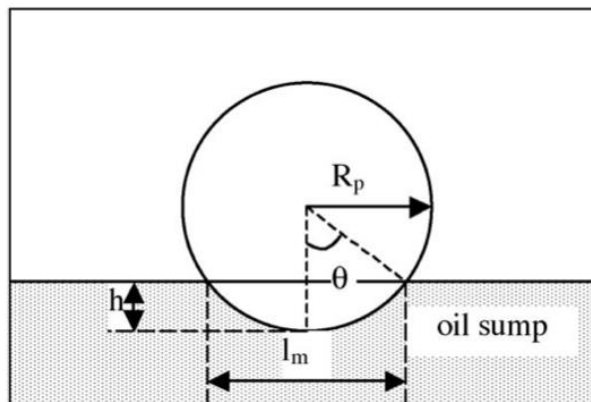
The lubricant used for the experiments was a common Automatic Transmission Fluid

Table 2 Lubricant properties

Kinetic viscosity at 40 degrees (Cst)	Kinetic viscosity at 100 degrees (Cst)	Density at 15 degrees (kg/m ³)
30	7.48	871

Experiment parameters:

- Gear 1: $h/R_p=0.5$
- Gear 2: $h/R_p=0.25$ & $h/R_p=0.5$





Improvements

- End plates remanufactured with higher precision for more accurate results
- Swivel joint to eliminate torsional stress from pipe
- New techniques used for data collection, higher precision
- 10 trials for each experiment, data collected at different rotational speed

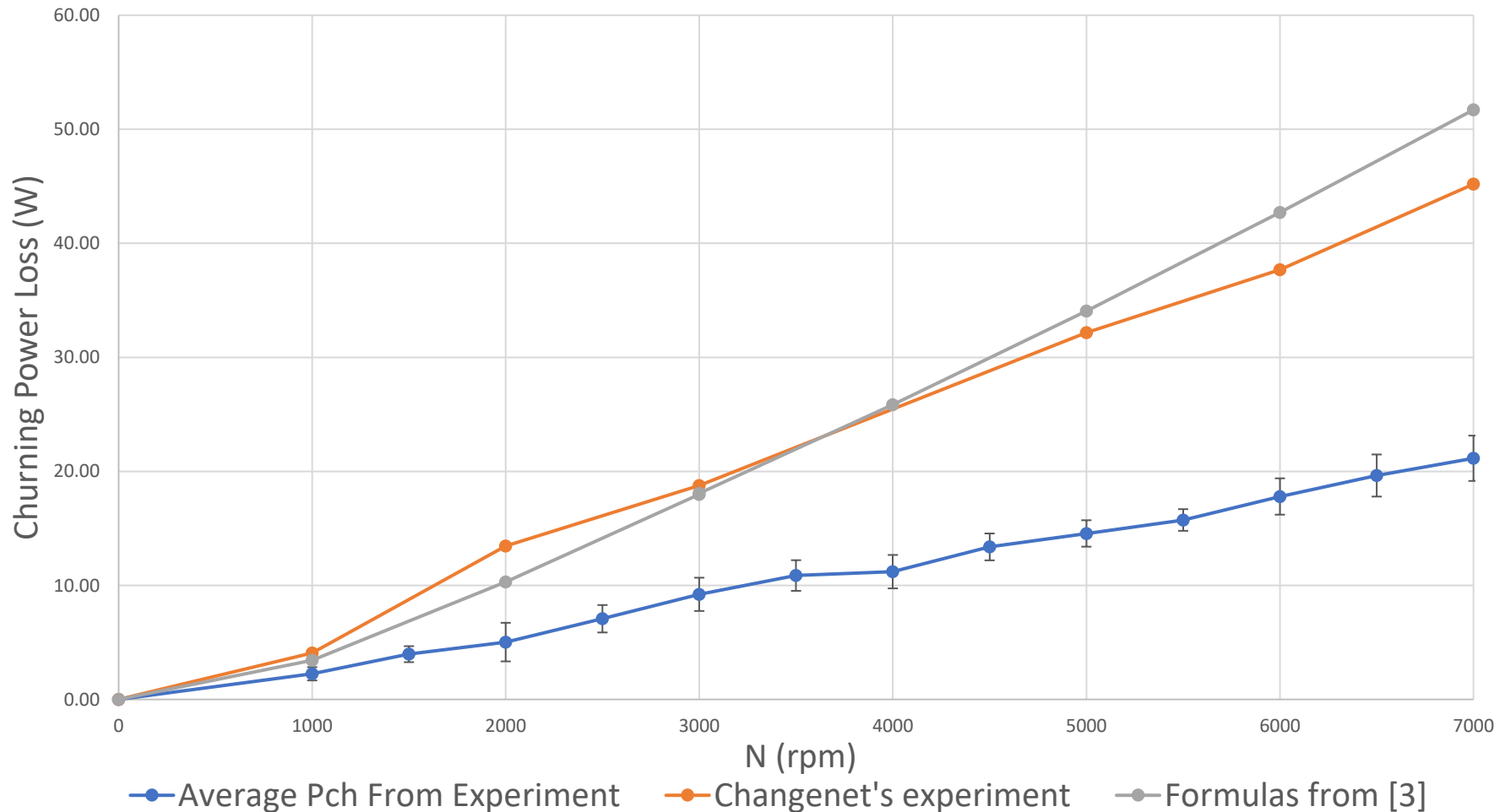




Data Analysis – Gear 1

Churning Power Loss Comparison

Gear 1, $h/R_p=0.5$

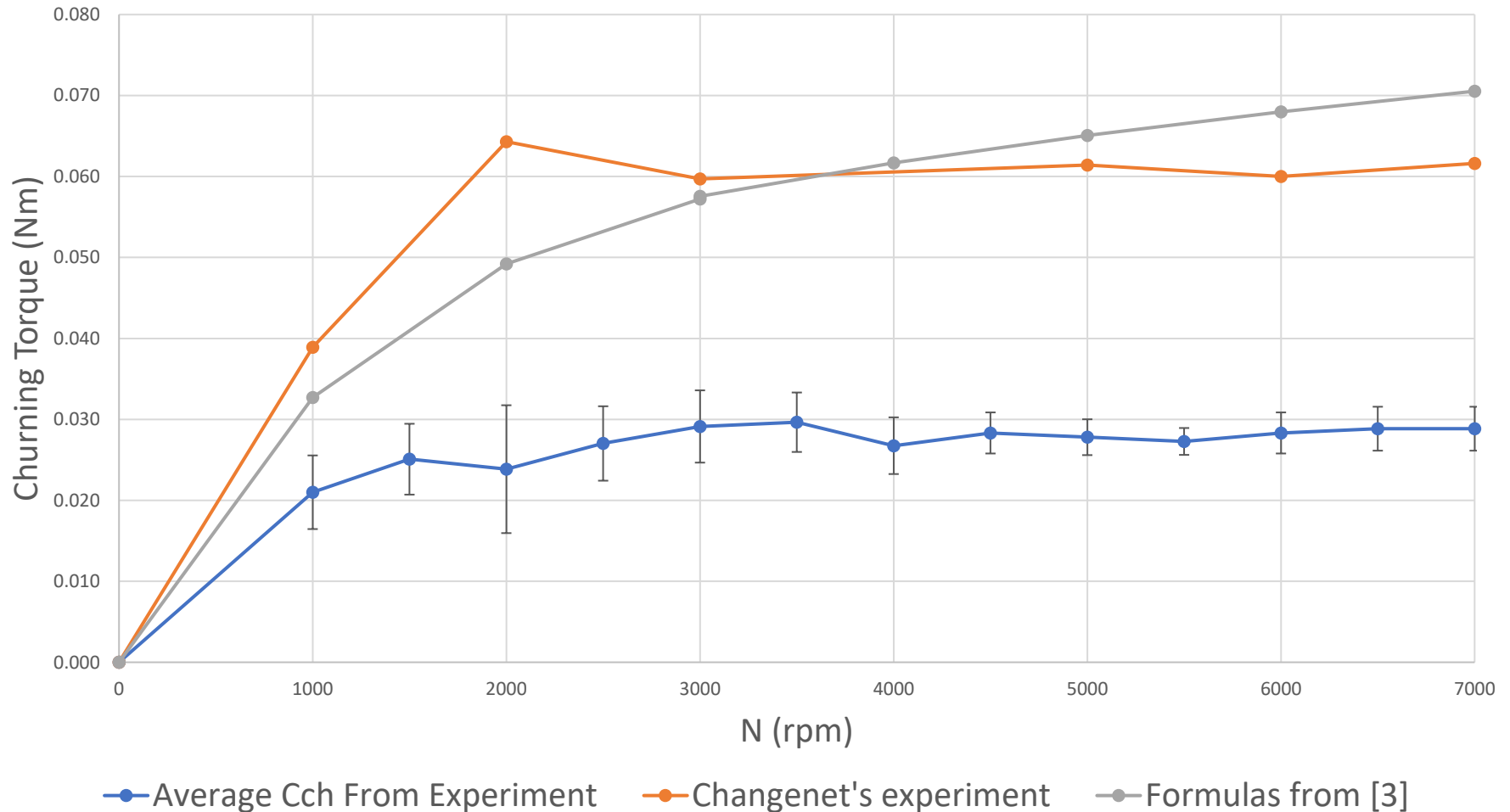


*128 data point



Churning Torque Comparison

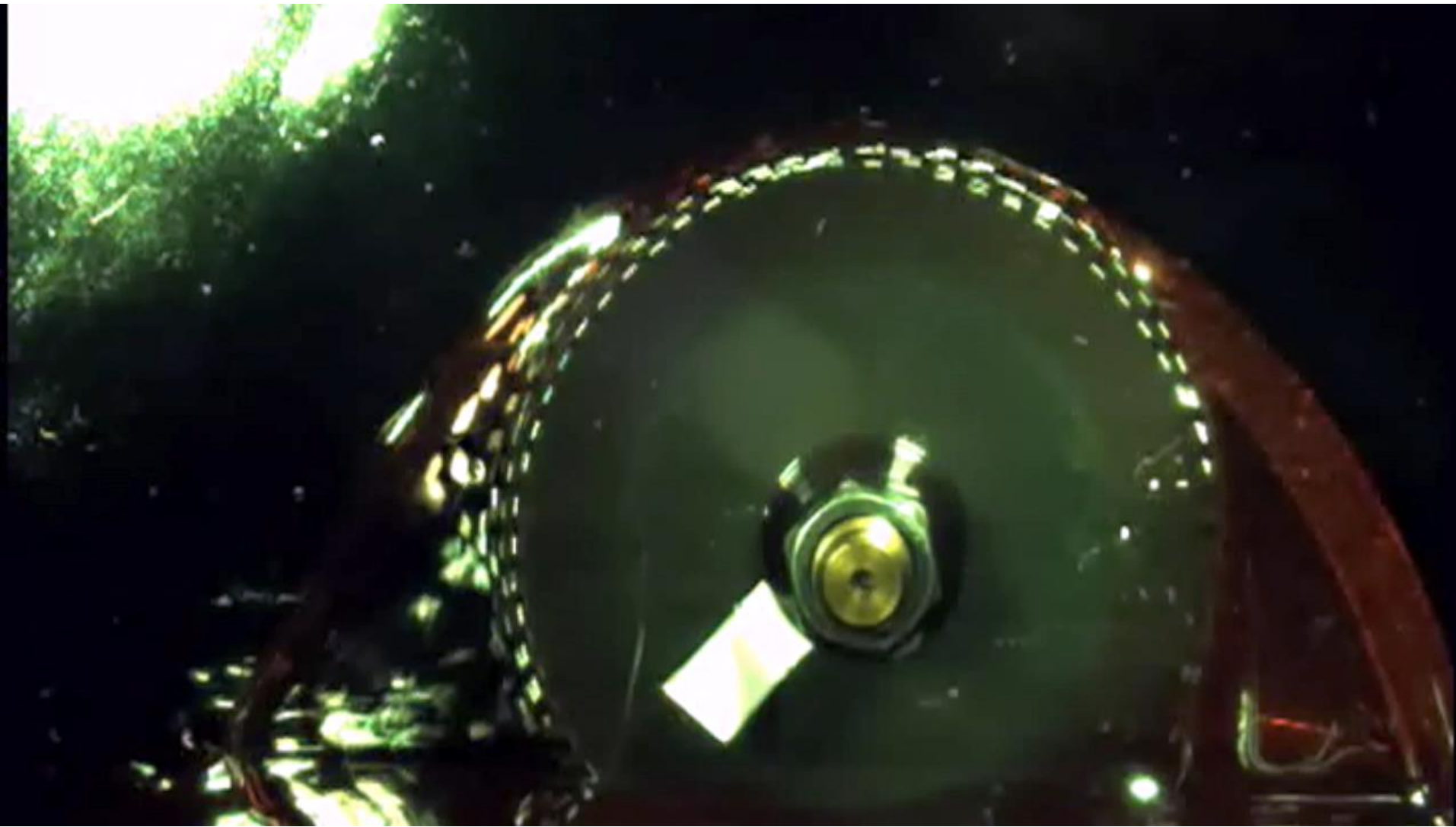
Gear 1, $h/R_p=0.5$





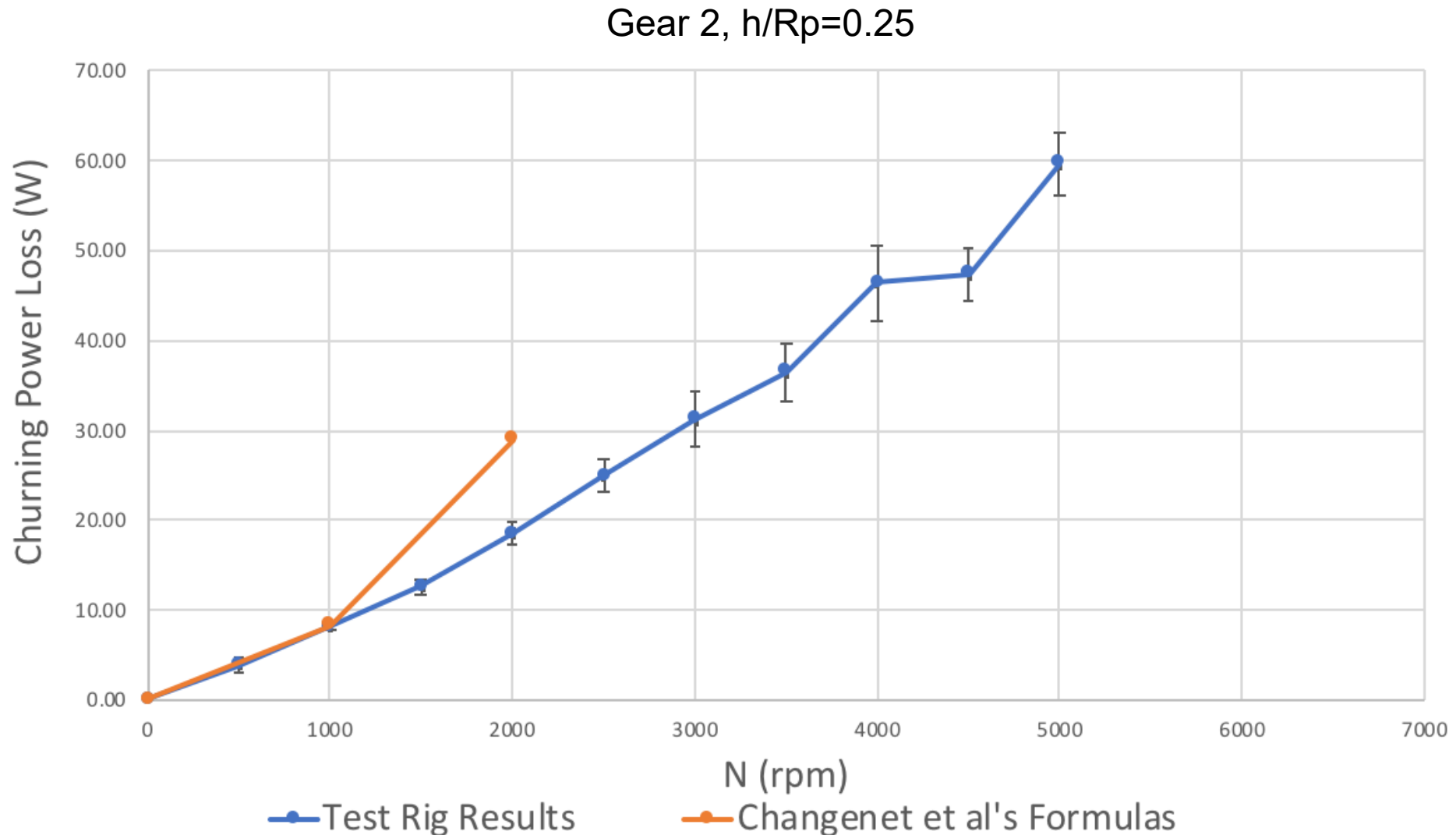
Flow regimes – Gear 1

Video of the splashing pattern for Gear 1; $h/R_p=0.5$



Data Analysis – Gear 2

Churning Power Loss Comparison

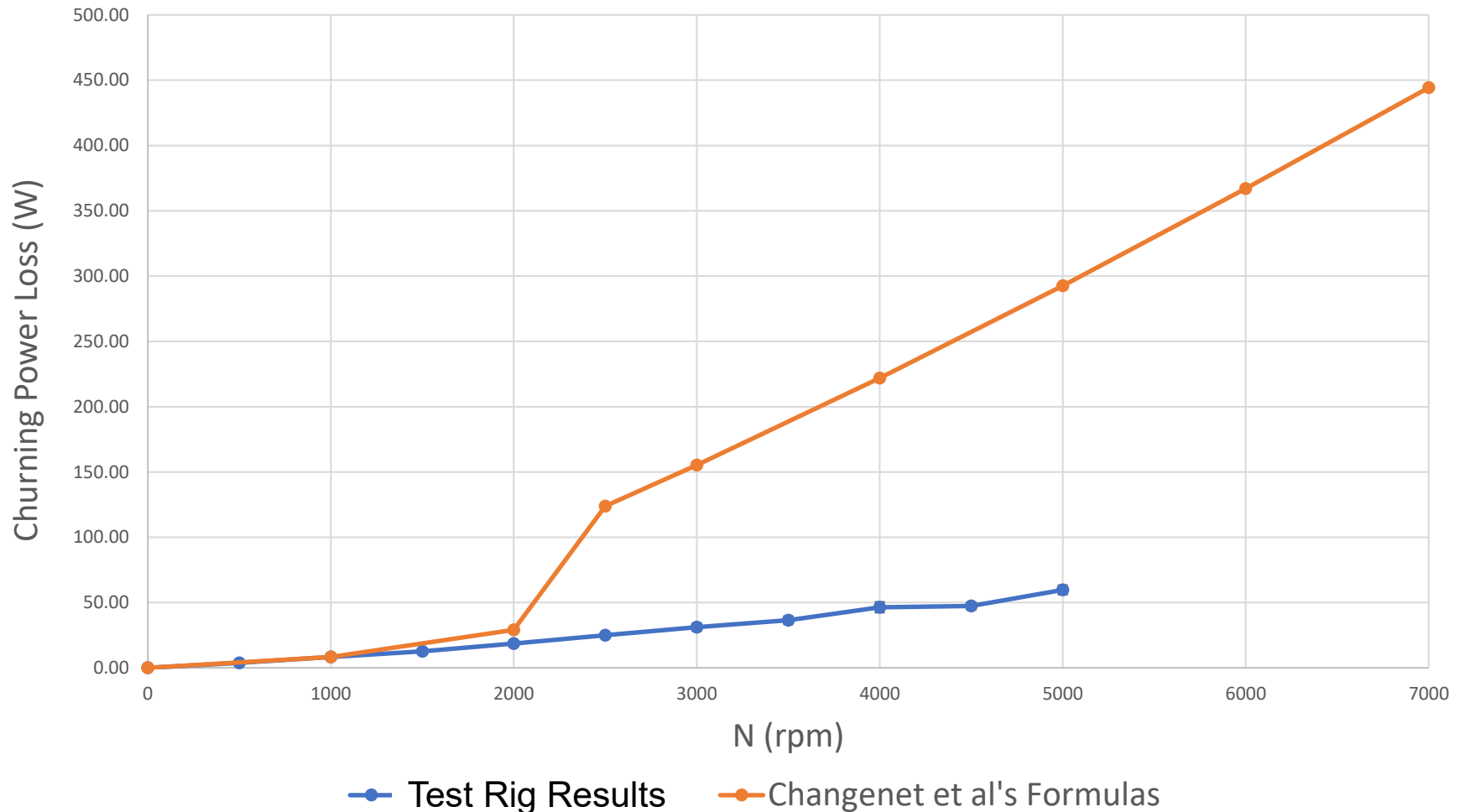




Data Analysis - Gear 2

Churning Power Loss Comparison

Gear 2, $h/R_p=0.25$

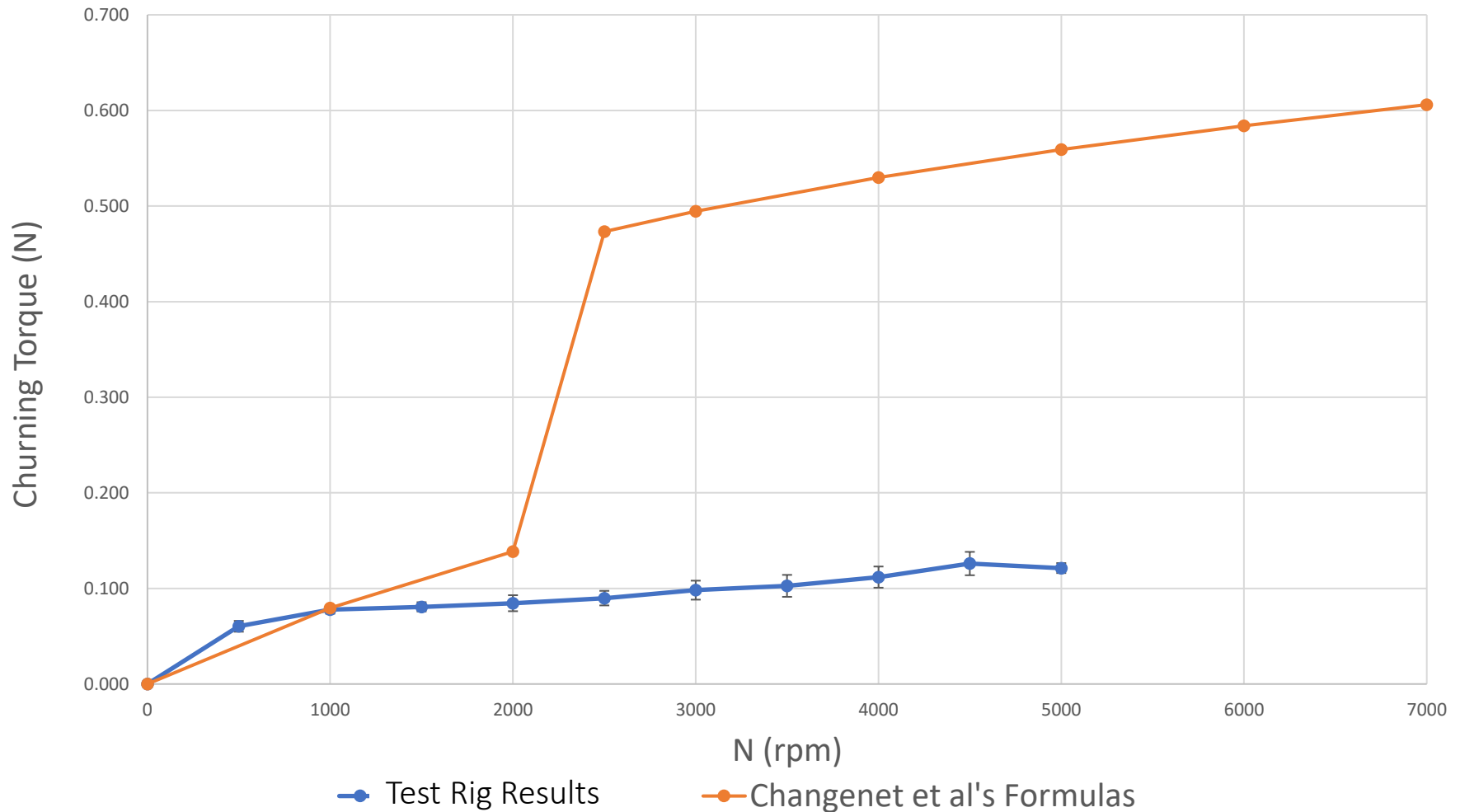




Data Analysis - Gear 2

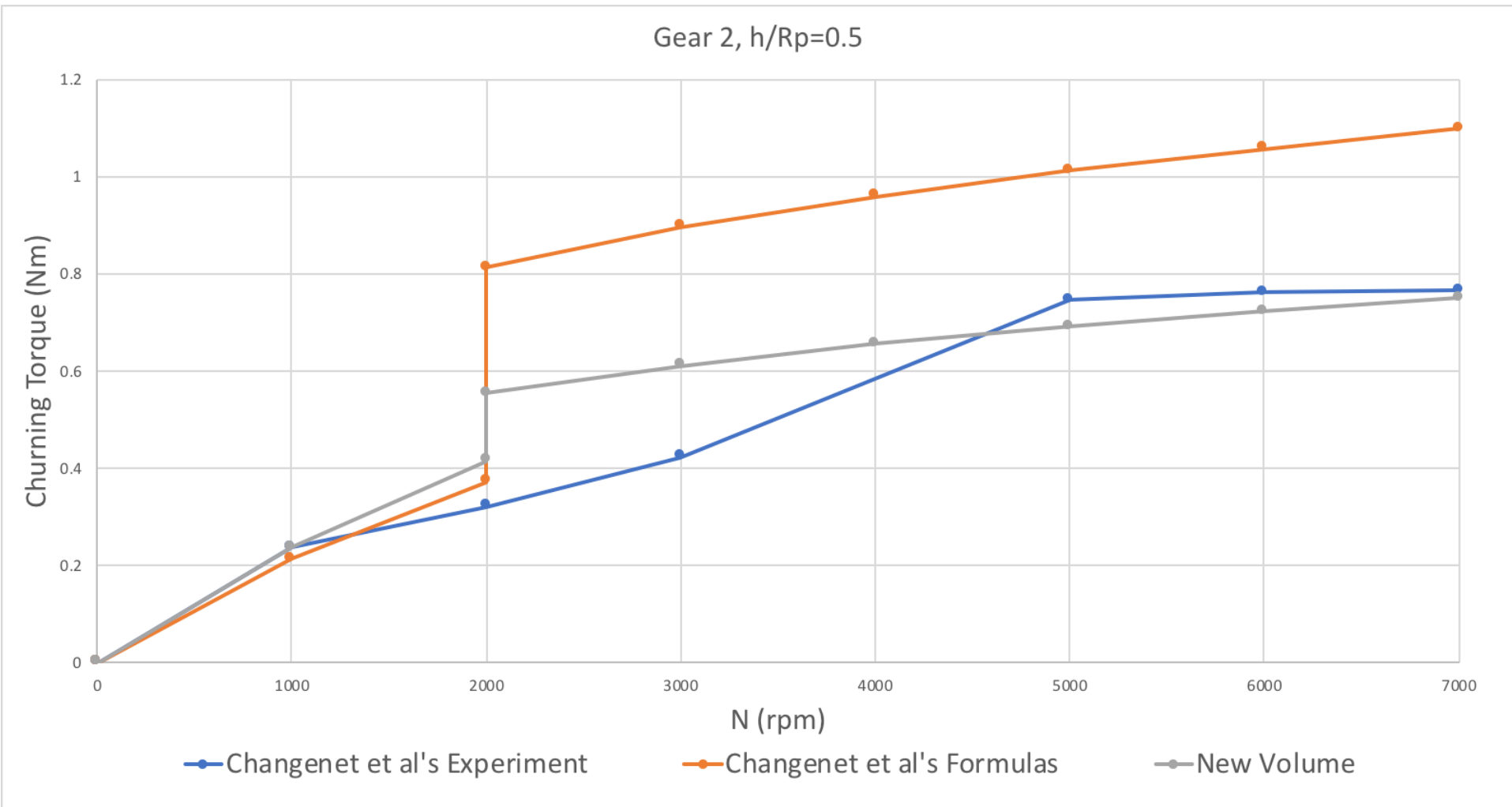
Churning Torque Comparison

Gear 2, $h/R_p=0.25$





Validating Formula



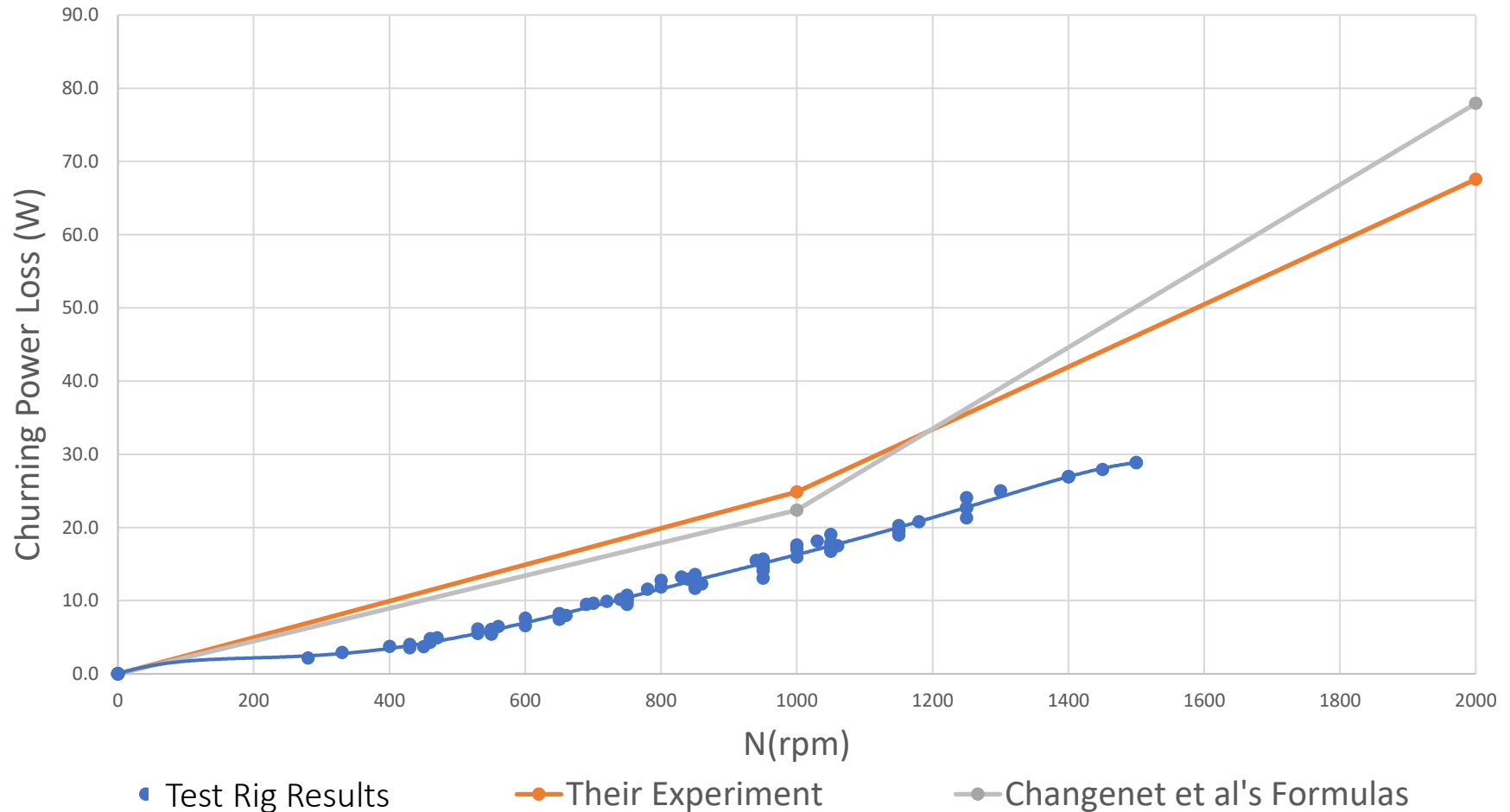
Orange line uses a smaller volume, grey line uses a greater volume



Comparison of Data Cont.

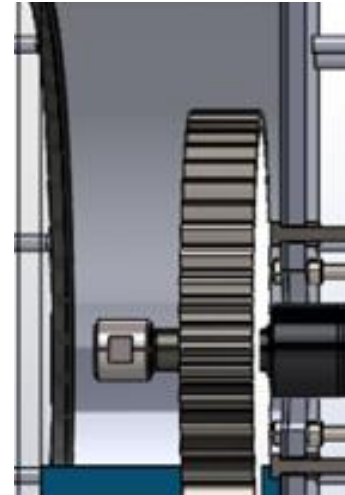
Churning Power Loss Comparison

Gear 2, $h/R_p=0.5$



Cause of Deviation

- The immersion depth decreases when in operation
- Gearbox Geometry,
 - Radial Vs Rectangular test rig
 - Axial Clearance (<5% at current range, 11mm)
- Higher percentage uncertainty for smaller gear
 - 19% for Gear 1 at 7000 rpm
 - 5% for Gear 2 at 5000 rpm
- Power loss due to heat transfer was noticeable but should not affect results
- None of which explains the huge difference in Gear 2



Cause of Deviation Cont.

- Formula used suggest that churning torque increase as volume decrease for $\gamma > 1250$ m/s, recall formula from earlier,

if $\gamma > 1250$ m/s² and $Rec > 4000$

$$C_m = 3.644 \left(\frac{h}{D_p} \right)^{0.1} \left(\frac{V_0}{D_p^3} \right)^{-0.35} Fr^{-0.88} \left(\frac{b}{D_p} \right)^{0.85}$$

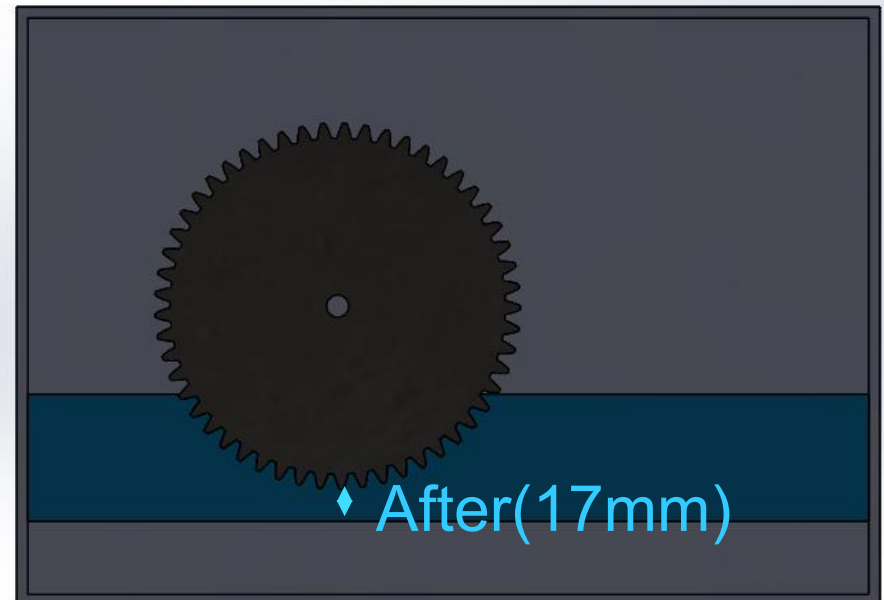
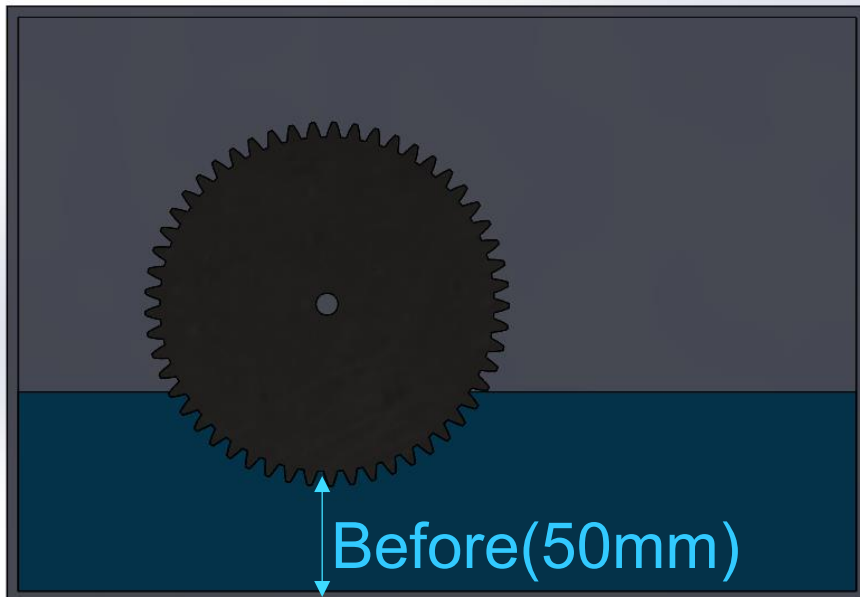
- Which suggest that:

$$C_{ch} \propto \frac{V_0^{-0.35}}{D_p^3}$$

- It was believed that due to the shape of the gearbox, it could be situational
- A simple model was made to recreate their gearbox to provide a closer view on their method on obtaining the coefficient shown above

Speculation

- The method used to reduce the volume was by inserting metal sheet
- However, this also decreases the depth
- Depth from pitch of gear to casing,
Before 50mm
After <20mm
- Choke point created, possibly interfering the flow regimes





Test Rig Limitation

- Mild leakage at several locations when in operation
- Oil level exceeding $h/R_p=0.5$ is not recommended
- Larger gears at high-speed cause too much splashing, Tachometer no longer liable
- Air-motor pressure inlet limit reached(3 bar) at ~2500 rpm for Gear 2
- Current newton meter restrict maximum torque measured ~0.2Nm
- No way of measuring internal temperature

- Gear 1 shows encouraging data and similar physical attribute to other study
- Churning torque leveled after certain speed, making
$$P_{ch} \propto \omega$$
- Testing range limited by design
- Questionable influence on the volume in Changenet's equations



Reference

- [1] Boness, R.J.1989. Churning Losses of Discs and Gears Running Partially Submerged in Oil. In: *Proceedings of the International Power Transmission and Gearing Conference, 25/28 April 1989, Chicago*. New York: American Society of Mechanical Engineers, pp.355-359.
- [2] Changenet, C., and Velez, C., 2007 “A Model for the Prediction of Churning Losses in Geared Transmissions-Preliminary Results,” vol. 129, no. (1), pp. 128-133.
- [3] Changenet, C., Leprince, G., Ville, F. and Velez, F. 2011. A Note on Flow Regimes and Churning Loss Modelling. *Journal of Mechanical Design*, 133, 121009.
- [4] Changenet, C., and Velez, P., 2008, “Housing Influence on Churning Losses on Geared Transmissions” *ASME J.Mech. Des.*, **130**(6), p.6



Question Time

Thank you for your time and feel free to ask any question regarding the project